Transformers, made clear.

A focused starter for electrical fundamentals used in SSC JE and RRB JE preparation. This is not a complete transformer chapter or syllabus guide.

01 / What fixes the core flux?

An alternating core flux induces an emf in each winding. For sinusoidal flux with maximum value Φ_m, the RMS induced emf is:

$$E = 4.44 f N \Phi m$$

Here E is in volts, f in hertz, N is the number of turns, and Φ_m in webers. The factor 4.44 assumes sinusoidal flux. With a small primary impedance drop, $V_1 \approx E_1$, so flux is approximately proportional to V_1/f.

02 / Ideal turns, voltage and current ratios

$$a = N1/N2 = V1/V2 = I2/I1$$

These are magnitude relations for an ideal transformer under load. Input power equals output power. Real transformers have winding losses, core losses, leakage flux and a nonzero excitation current.

03 / The no-load current has two components

In the usual approximate model, I_c is in phase with supply voltage and supplies core loss. I_m is approximately 90° behind voltage and establishes flux. Thus $I_0 = \sqrt{(I_c^2 + I_m^2)}$, not $I_c + I_m$.

When load increases at fixed voltage and frequency, the primary draws an additional load component. It largely balances the secondary ampere-turns; it does not mean that core flux rises in proportion to load.

When conditions change.

Use the assumptions before using the formula. These comparisons assume the same winding turns.

Voltage decreases; frequency fixed

Core flux decreases approximately with voltage. In the normal unsaturated operating region, magnetising current generally decreases. It does not rise to restore the old flux.

Frequency decreases; voltage fixed

Core flux increases approximately as 1/f. Saturation and a large excitation current can result. Do not assume magnetising current is linear with flux near saturation.

Voltage and frequency decrease together

If V/f is held constant, core flux is approximately constant while winding drops are small. At very low frequency, winding resistance makes the approximation less accurate.

Load increases; voltage and frequency fixed

Primary load current increases. Core flux stays approximately constant. Copper loss increases approximately with the square of load current at fixed resistance.

Power factor decreases; kVA fixed

Current is unchanged at fixed voltage, so copper loss is approximately unchanged. Real output power decreases. Do not confuse fixed kVA with fixed kW.

Solve it without a calculator.

Three original worked examples. These are practice models, not actual previous-year questions.

01 / Voltage, frequency and flux

A transformer operates at 200 V and 50 Hz. The supply changes to 180 V and 45 Hz. Neglect primary winding drops. Find the new-to-old maximum flux ratio.

Φnew/Φold = (180/200) × (50/45) = 1

Both voltage and frequency fall by 10%. Their ratio is unchanged, so the maximum flux is approximately unchanged.

02 / No-load current

The core-loss component is 0.6 A and the magnetising component is 0.8 A. Assume these components are perpendicular. Find no-load current and no-load power factor.

I0 = √(0.6² + 0.8²) = 1 A; pf = 0.6

Recognise the 3-4-5 triangle scaled by 0.2. The power factor is I_c/I_0, and is lagging in this model.

03 / Load for maximum efficiency

Core loss is 100 W and full-load copper loss is 400 W. At fixed voltage, frequency, resistance and load power factor, find the fraction x of full load for maximum efficiency.

x² × 400 = 100 → x = 1/2

Maximum efficiency occurs when variable copper loss equals approximately constant core loss. The answer is half load, not one-quarter load. This is ordinary efficiency, not all-day energy efficiency.

Try it. Then check it.

Six original practice questions with short solutions. Cover the answer section while attempting them.

1. An ideal transformer has $N_1/N_2 = 5$. Its primary voltage is 230 V. What is its secondary voltage?

2. For that ideal transformer, secondary load current is 10 A. What is primary current?

3. At fixed frequency, applied voltage falls by 20%. Neglect winding drops. How does maximum flux change?

4. At constant winding resistance, load current is halved. What fraction of the original copper loss remains?

5. $I_0 = 5$ A and $I_{□} = 3$ A in the approximate no-load model. Find I_m.

6. Core loss is 200 W and full-load copper loss is 800 W. At what load fraction is ordinary efficiency maximum under the usual constant-loss assumptions?

Answers and reasoning

1. 46 V. $V_2 = 230/5$.

2. 2 A. $I_1 = I_2/a = 10/5$.

3. It falls by approximately 20%, since $\Phi_m \propto V/f$.

4. One-quarter. Copper loss $\propto I^2$.

5. 4 A. $I_m = \sqrt{(5^2 - 3^2)}$.

6. Half load. $x = \sqrt{(200/800)} = 1/2$.

Next topics to study: equivalent circuit, open-circuit and short-circuit tests, regulation, losses, efficiency, autotransformers and parallel operation.